



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 06.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: CE G547
Course Title	: Pavement Failure Evaluation and Rehabilitation
Instructor-in-Charge	: Nishant Bhargava
Instructor(s)	: --
Tutorial/Practical Instructors	: --

1. Course Description

This course covers topics related to pavement failure, evaluation and rehabilitation such as pavement deterioration, distress and different types of failures, methods of measurement and treatment, use of modern equipment for pavement surface condition measurements, analysis of data, interpretation, Structural deterioration of pavements: causes, effects, methods of treatment, structural evaluation of flexible pavements by rebound deflection method, analysis of data, design of overlay, use of FWD and other methods for evaluation of flexible and rigid pavements and their application, pavement testing under controlled conditions, test track studies, Instrumentation for pavement testing.

2. Scope and Objective of the Course

The course includes the determination of causes of failure in flexible and concrete pavement. The different techniques for evaluation and rehabilitation of different types of distresses will be taught to the students. The use of modern equipment's for structural evaluation of pavement, and design of overlay with the help of data collected by using the equipment will be covered in this course. The accelerated test facilities constructed worldwide and instrumentation used in these test facilities for pavement evaluation will be taught to the students.

3. Text Books

T1 : Huang, Y.H., *Pavement Analysis and Design*, Pearson, 2004.

Reference Books

- R1: Mallick R. B. and El-Korchi T. *Pavement Engineering: Principles and Practice*, CRC Press.
- R2: Pearson D. *Deterioration and Maintenance of Pavements*, Institution of Civil Engineers.
- R3: Relevant Federal Highway Administration Reports and NCAT Reports
- R4: Relevant IRC codes



4. Course Plan

Module No.	Lecture Session	Reference	Learning outcomes
1-10	Pavement deterioration, distress and different types of failures, methods of measurement and treatment	T1, R2, R3	Identification of distresses, its possible causes, measure its severity and suggest adequate treatments.
11-14	Use of modern equipment for pavement surface condition measurements, analysis of data, interpretation	T1, R3, R4	Pavement condition survey to quantify the extent of distresses and propose suitable maintenance or rehabilitation requirements.
15-33	Structural deterioration of pavements: causes, effects, methods of treatment	R4	Structural evaluation of pavement
18-33	Structural evaluation of flexible pavements and design of overlay	R4	Understand the functioning of state-of-the-art equipment and interpret field data to derive input parameters for designing pavement overlays.
34-40	Pavement testing under controlled conditions, accelerated testing and evaluation methods, test track studies and instrumentation for pavement testing	T1, R3	Advance pavement testing through innovative methodologies and instrumentation.

5. Evaluation Scheme

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 min.	25	05/10/2026 AN2	CB/OB
Comprehensive Examination	3 hr.	35	03/12/2026 AN	CB/OB
Quizzes	45 min.	10	*	OB
Laboratory	--	10	*	OB
Project	-	20	*	OB

* shall be announced prior

6. Chamber Consultation Hour

Tuesday 3 pm – 4 pm (1212 – F)

7. Notices

Notices of different components of evaluation scheme will be sent by email.



8. Grading Policy

- If the student absents herself/ himself even in one of the evaluative components (listed in the evaluation scheme) entirely, his / her performance shall be reported as NC (Not Cleared) regardless of his/ her final total score in the course (See Clause 4.19 of BITS Academic Regulations). Students gaining less than 10% overall, may be awarded NC grade.
- Project grading: The student is expected to conduct presentation for the project. If absent for presentation, the student will be deemed as absent for the evaluative and will be awarded NC grade.

9. Make-up Policy

- Make up will be granted only on genuine reasons and will be considered on a case by case basis. However, prior permission is a must. It should follow the official request procedure. No other mode of request shall be entertained.
- For medical cases, a certificate from the concerned physician from the Medical Centre must be produced (Subjected to Academic regulations).
- Make up policy is only applicable to Mid – Semester test and Comprehensive Examination. It is not applicable to quiz and project presentation.

10. Note

Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

Instructor-in-charge
Course No. CE G547